

CUSUM RAP Tool - Developer and Administrator Guide

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CUSUM RAP Tool: Developer & Administrator Guide

This guide is intended for developers, IT administrators, and maintainers of the CUSUM RAP Tool. If you are an end-user looking for instructions on how to use the analytical interface, please refer to the **User Manual** located at www/docs/Documentation.pdf or click the “Help” button inside the running application.

1. Project Structure

The repository is structured as a standard Shiny application with modularized components.

- **Root Directory (/):** Contains the main application scripts (app.R, global.R), project files (CUSUM-RAP.Rproj), and top-level configurations (config.json).
- **R/ Directory:** Contains the core logic and Shiny modules. This includes mod_cusum.R (the main UI and Server module) and the backend scripts (functions_cusum.R, functions_io.R, functions_parameters.R, functions_plot.R).
- **www/ Directory:** Contains static web assets exposed to the browser. This includes config.css (UI styling), the docs/ folder (end-user documentation), and graphical assets (AnalysisforAction_white.png, splash.png, default.ico).

```
/ (Root Directory)
├── app.R                # Main application script
├── global.R             # Configuration and global variables script
├── CUSUM-RAP.Rproj      # RStudio project file
├── config.json           # Top-level configurations (e.g., spatial hierarchies)
├── R/                   # Core logic and Shiny modules
│   ├── mod_cusum.R      # Main module (UI and Server)
│   ├── functions_cusum.R # Mathematical algorithm functions
│   ├── functions_io.R   # Data reading and saving functions
│   ├── functions_parameters.R # Variable calculation functions
│   └── functions_plot.R  # Chart and plot generation functions
├── www/                 # Static web assets (exposed to the browser)
│   ├── config.css       # Stylesheet (UI colors and layout)
│   ├── AnalysisforAction_white.png # Graphic asset
│   └── splash.png        # Graphic asset
```

```

├── default.ico           # Browser tab icon
└── docs/
    └── Documentation.pdf # The manual that opens when clicking "Help"

```

2. Running the App Locally (Development)

To run or test the application within an R development environment: 1. Open the CUSUM-RAP.Rproj file in RStudio. 2. Open app.R. 3. Click **Run App** in the RStudio IDE, or run `shiny::runApp()` in the console.

3. Web Deployment

To make the tool accessible via a web browser, you must host the application on a server.

Option A: shinyapps.io (Easiest Cloud Hosting) 1. Create a free or paid account at shinyapps.io. 2. Install the deployment package in R by running `install.packages("rsconnect")`. 3. Authenticate your RStudio session using the secure token provided in your shinyapps.io dashboard. 4. Deploy the app by clicking the blue **Publish** button in the top right corner of your RStudio script editor, or by running `rsconnect::deployApp()` in the console.

Option B: Self-Hosted Infrastructure If your epidemiological data requires strict privacy compliance that prevents the use of third-party cloud hosting, you can host the application on your own infrastructure (AWS, DigitalOcean, or an internal secure server). This requires installing the open-source **Shiny Server** on a Linux machine or containerizing the application using **Docker**.

4. App Configuration & Customization

Administrators can customize the application's appearance and geographical mapping without altering the core R logic.

- **Modifying the UI Design (www/config.css):** The application's styling is hardcoded into this CSS file. Find and replace specific hex codes to change branding (e.g., #4a4a4a for dark gray backgrounds, #616161 for standard text).
- **Adding Countries & Regions (config.json):** The spatial hierarchy for the analytical output is defined here. Add a new array defining the administrative levels from highest to lowest (e.g., "Brazil": { "levels": ["country", "state", "municipality"] }).